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Lung: Research

Achieving Equitable Care for Racial Minority Patients With a Lung Cancer Screening Program

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ABSTRACT

BACKGROUND Racial minority patients with lung cancer have more advanced stage, lower resection rates, and worse overall survival, especially among Black men. Lung cancer screening with low-dose computed tomography (LDCT) can reduce mortality; however, the influence of race on access is unknown. This study evaluated LDCT's impact on timely diagnosis and treatment.

METHODS Patients undergoing LDCT at an urban academic center were enrolled in a cohort study from November 2012 to February 2019. Baseline demographic, cancer-specific, and treatment-specific factors were analyzed and compared with city and state averages. Subgroup analysis was performed by sex.

RESULTS The study cohort included 2552 patients (male, 51.3%; median age, 63 years; White, 70.4%; Black, 28.6%; average smoking pack-years, 43). Higher screening rates for Black men (12.9% vs 6.11%; $P < .0001$) and Black women (15.7% vs 6.68%; $P < .0001$) were found compared with the greater metropolitan area population. Lung cancer was identified in 3.7% of the screened population. Lung cancer incidence was similar between Black and White patients (3.3% vs 3.9%; $P = .46$), with no difference in stage at diagnosis. There were increased rates of early-stage diagnosis compared with state averages (52.5% vs 27.0%; $P < .0001$). Black patients had no differences in delays to care (>8 weeks) and equitable rates of surgical resection compared with White patients (63.6% vs 62.9%; $P = .95$).

CONCLUSIONS A structured LDCT program can provide access to lung cancer screening to a high number of racial minority patients and provide equitable care through timely diagnosis and treatment. This study underscores the importance of optimizing screening to deliver equitable outcomes for all lung cancer patients.

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In the United States, lung and bronchus cancer is the third most common new cancer diagnosis, responsible for 49.1 cancers in 100,000 people.[1] Lung cancer is also the leading cause of cancer mortality.[2] These statistics are complicated by pervasive racial and ethnic disparities, with Black men having both higher incidence and higher mortality rates.[3,4] Among Black men, this results in 71 new cases per 100,000 compared with 64 new cases per 100,000 in their White counterparts.[5]

Lung cancer screening results in a 20% to 30% reduction in cancer mortality by detecting lung cancer at earlier stages.[6] However, the American

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Lung Cancer Association reports that only 4.5% of eligible patients are screened nationally. Additionally, there are major disparities in lung cancer screening use among racial minorities.[7-9]

Secondary analyses of the National Lung Cancer Screening Trial have demonstrated that these disparities extend to differences in treatment and survival among those patients enrolled in the clinical trial. Black men are less likely to undergo lung resection and have worse lung cancer survival.[3,10,11] Additionally, they are more likely to have a diagnosis of advanced disease and less likely to receive the necessary treatment.[6,12,13] Current literature has explored the contribution of smoking patterns to these inequities, but there has not been adequate consideration of screening-related interventions.[14]

There remains a gap in understanding the utility of establishing a structured low-dose computed tomography (LDCT) screening program to provide equitable care to racial minority patients in real-world clinical practice.[15] The purpose of the study was to define the impact of a structured LDCT lung cancer screening program on providing timely diagnosis and treatment along the continuum of patient care and to evaluate for any disparities between racial groups.

MATERIAL AND METHODS

This retrospective cohort study was performed at the University of Cincinnati after Institutional Review Board approval (IRB #2018-7564, approved on December 19, 2018). The University of Cincinnati Lung Cancer Screening program enrolled patients undergoing lung cancer screening with LDCT from November 2012 to February 2019. The study's target population consisted of patients aged 55 years and older who underwent LDCT and who had a new computed tomographic (CT) finding concerning for lung cancer that prompted further evaluation by a multidisciplinary tumor board. The LDCT screening program was accessible to any patient who met US Preventive Services Task Force lung cancer screening guidelines. Patients were referred by any physician or provider in the greater Cincinnati area or who self-referred through the website. The lung cancer screening program is run by full-time nurses and administrative support staff. The medical director of the program (S.L.S.) works closely with this team and other members of the working group (S.K., R.V.H.). The program was promoted to primary care clinics during implementation, and these marketing activities are continuing. Study exclusion criteria included any patients with a confirmed case of lung cancer before the start of the study period. The primary outcome of interest was incident lung cancer, as defined by the International Classification of Diseases, 10th Revision (ICD-10) diagnosis code C34 in the patient's electronic medical record.

The institution's electronic medical record was used to collect demographic information, residential mailing address, age, race, sex, and patient status (alive or deceased); diagnoses were defined according to ICD-10. To collect LDCT screening data, any completed procedures listed under the "LDCT" code were extracted. Any "no show," "incomplete," or "cancelled" LDCT screenings were excluded. Smoking history (current or former smoker, years since quitting, pack-years and pack per day history), respiratory risk factors including chronic obstructive pulmonary disease (COPD) and pulmonary fibrosis diagnoses, and history of cancer were recorded. Insurance type was collected as public (Medicaid and Medicare), private (commercial), both (combination of public and private), and other (anyone who self-paid, had worker's compensation or another governmental or nongovernmental insurance).

Additionally, the following were recorded: staging of the suspected lung cancer on the basis of the seventh edition of the *AJCC [American Joint Committee on Cancer] Cancer Staging Manual*[16]; the type of lung cancer; and the intervention type, defined as definitive therapy through surgical resection, chemotherapy, radiation, or a combination of therapies. The institution's lung cancer screening program comprises 3 nurse coordinators, and all Lung RADS (Lung Imaging Reporting and Data System) 4 nodules are presented to the thoracic tumor board. Data on the time (days) from diagnosis to specialist referral, the time (days) to bronchoscopy or biopsy, the time (days) to an intervention, the type of intervention or treatment, and survival were also collected. A cutoff of 8 weeks from the initial CT finding to intervention was used to define delayed treatment.[17,18]

Deprivation indexes were calculated for both groups by using 6 different variables: (1) the fraction of the population with income below the poverty level; (2) the median household income; (3) the fraction of the population aged 25 years

TABLE 1 Patient Demographics and Clinical Characteristics

Factors	Screening Cohort, n = 2552	Lung Cancer Cohort, n = 90	Surgical Resection Cohort, n = 57
Age, y	63 [59–68]	67.2 [61.3–70.8]	67.4 [61.2–70.3]
Sex, male	51.3	42.7	45.3
Race			
White	70.4	74.0	73.6
Black	28.6	26.0	26.4
Other	0.9	0	0
Insurance			
Public	55.7	68.1	68.6
Private	19.4	10.6	11.8
Combination	9.9	20.2	17.6
Other	4.9	1.1	2.0
Uninsured	0.2	0	0
Deprivation index	0.38 [0.27–0.50]	0.38 [0.27–0.50]	0.40 [0.28–0.53]
Smoking history			
Current smoker status	54.5	53.2	49
Average pack-years	46 [38–60]	50 [40–72.2]	45 [39–67.5]
Pulmonary risk factors			
COPD	36.5	48.9	45.1
Pulmonary fibrosis	5.4	9.6	11.8
History of cancer	15.5	14.9	21.6

Values are median [interquartile range] or %. COPD, chronic obstructive pulmonary disease.

and older with an educational attainment of at least high school graduation; (4) the fraction of the population with no health insurance coverage; (5) the fraction of households receiving public assistance income, food stamps, or Supplemental Nutrition Assistance Program; and (6) the fraction of houses that are vacant.[19,20] Each of the 6 variables contributed equally to the calculation of the ZIP Code-based deprivation index. This value is quantified on a scale of 0 to 1 and is based on Census-tract-level community data derived from the 2015 American Community Survey. Population data from the greater Cincinnati metropolitan area was compared with the study's screening cohort. Finally, state-level data on lung cancer incidence, rates of early-stage diagnosis (stage I), and stage at diagnosis from the American Lung Association's 2023 Ohio data were collected and compared.[5]

Continuous variables were described as mean (SD) for normal distributions and median [interquartile range (IQR)] for nonparametric distributions and were compared using analysis of variance and the Wilcoxon rank sum test as appropriate. Categorical variables were described as percentage and number and were compared using the χ^2 and Fisher exact tests. Statistical analyses were performed using SAS software version 9.4, JMP Pro 17 (SAS Institute), and significance was determined as $P < .05$.

RESULTS

The cohort included 2552 patients with a median age of 63 years [IQR, 59-68 years] (Table 1). Of these patients, 51.3% were male, 70.4% were White, and 28.6% were Black (Figure 1). The cohort when stratified by race and sex consisted of 37.8% White men, 32.6% White women, 12.9% Black men, and 15.7% Black women (Figure 2). Insurance data collected revealed 55.7% public, 19.4% private, 19.9% combination of public and private, 4.9% other, and 0.2% uninsured. The deprivation index for the entire cohort was 0.38 [IQR, 0.27-0.50] (Table 1).

In the study, 54.5% of the cohort reported current smoker status, and 44.4% were former smokers. The average number of pack-years smoked was 46 pack-years [IQR, 38-60 pack-years]. Pulmonary risk factors were present in 41.9% of the cohort; 36.5% of these patients had a COPD diagnosis, and 5.4% had a pulmonary fibrosis diagnosis (Table 1).

When compared with the population in the Cincinnati greater metropolitan area, a demographic breakdown of the study cohort revealed an increased screening rate of both Black men (12.9% vs 6.11%; $P < .0001$) and Black women (15.7% vs 6.68%; $P \leq .0001$) (Figure 3). The screening rates were not significantly different among White men and women.

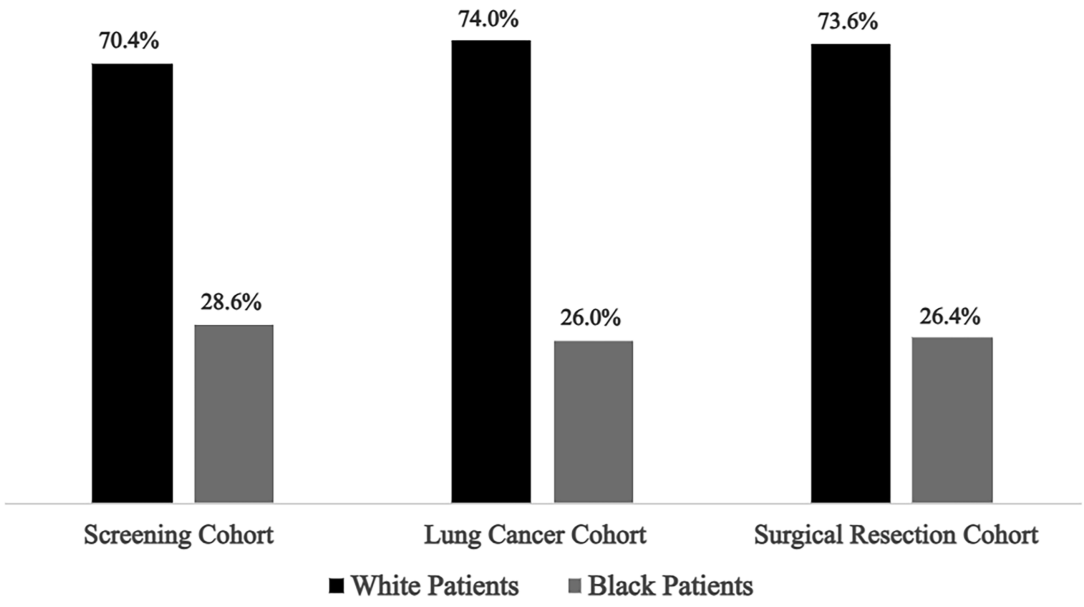


FIGURE 1 Comparative racial breakdown of cohorts demonstrating consistent percentiles across races: White patients (black shading); Black patients (gray shading).

Lung cancer was diagnosed in 90 (3.7%) patients. Most lung cancers were early stage (stage 1, 52.5%; stage 2: 23,8%; stage 3,15%; stage 4, 8.8%). A total of 63.5% of lung cancers were identified as adenocarcinoma, 22.4% as squamous cell carcinoma, 8.2% as small cell lung cancer, and 5.9% as other. There were no significant differences in staging or cancer type by sex, race, insurance status, or smoking status (Table 2). Of the patients with lung cancer, 42.7% were male, 74.0% were White, and 26% were Black (Figure 1). The cohort of patients with lung cancer when stratified by race and sex included 35.1% White men, 39.4% White women, 5.3% Black men, and 20.2% Black women (Figure 2). Insurance status included 68.1% public, 10.6% private, 20.2% combination of public and private, 1.1.% other, and 0% uninsured. In this lung cancer group, the average number of pack-years was 50 pack-years [IQR, 40-72.2 pack-years] with 1 average pack per day [IQR, 1-1.6 average pack per day]. A total of 14.9% of these patients had a history of cancer, 48.9% of these patients had a COPD diagnosis, and 9.6% had a pulmonary fibrosis diagnosis (Table 1). The incidence of lung cancer was similar between Black and White patients (3.3% vs 3.9%; $P = .46$) (Table 2). Of White men screened, 3.4% had a lung cancer diagnosis, compared with 1.5% of Black men screened ($P = .08$). Black men were less likely to have a diagnosis of lung cancer compared with Black women (1.5% vs 4.7%; $P = .02$). When compared with Ohio state-level data, the LDCT screening cohort had significantly increased rates of early-stage diagnosis (52.5% vs 27%; $P < .0001$).

Of the 90 incident cases, 63.1% of patients underwent resection, 45.2% had chemotherapy, and 28.6% had radiation therapy, either concurrently (34.4%) or as monotherapy. In the surgical resection cohort, 69.8% underwent initial surgery, whereas 30.2% had neoadjuvant therapy followed by surgery. The average time from CT finding to specialist referral was 17 days [IQR, 9-32 days] (Figure 4). A total of 62% of patients were referred to a thoracic surgeon, 30.4% were referred to a pulmonologist, and 7.6% were referred to an oncology specialist for further management, with no significant differences in treatment between races. The average time from specialist referral to bronchoscopy/biopsy was 19 days [IQR, 8-42 days], and the average time from bronchoscopy or biopsy to recommended intervention was 11.5 days [IQR, 0-45.3 days] (Figure 4). For all steps in treatment, there were no significant differences in referral timing by sex, race, insurance status , or smoking status. A total of 60.2% of the cohort took greater than 8 weeks from the initial finding on CT to an intervention, with no significant demographic differences.

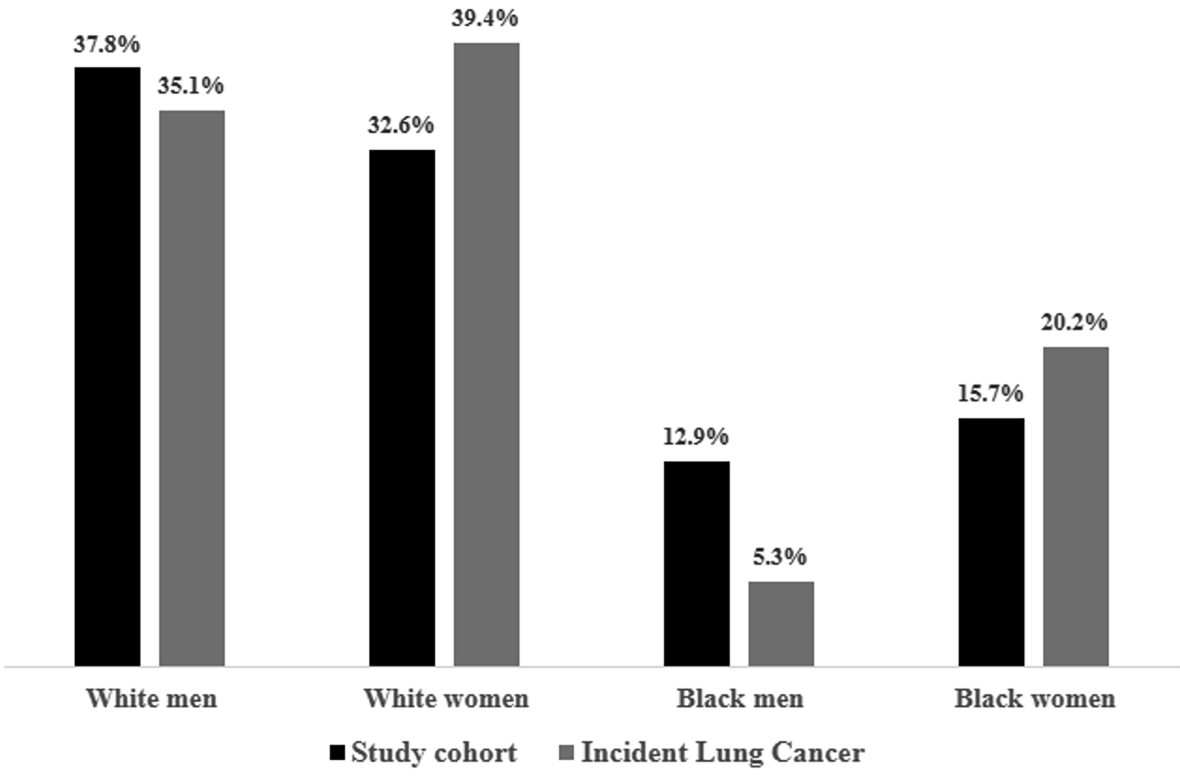


FIGURE 2 Study cohort screening population and lung cancer diagnoses stratified by race and sex: study cohort (black shading); incident lung cancer (gray shading).

Deprivation indexes calculated between the delayed group (>8 weeks) and those not delayed (<8 weeks) were not significantly different, at 0.37 and 0.41, respectively. There were also no significant differences in staging, intervention type, or survival between the delayed and not delayed cohorts.

In patients who underwent surgical resection for lung cancer (n = 57), the average age was 67.4 years [IQR, 61.2-70.3 years], 45.3% were male, 73.6% were White, and 26.4% were Black (Figure 1). Insurance status data collected revealed 68.6% public, 11.8% private, 17.6% combination of public and private, 2% other, and 0% uninsured. The average deprivation index was 0.4 [IQR, 0.28-0.53), with 49% having a current smoking status and an average of 45 pack-years [IQR, 39-50 pack-years] (Table 1). Further analysis of demographic subgroups revealed equitable resection rates between Black and White patients (63.6% vs 62.9%; $P = .95$). When further analyzed by sex, resection rates among racial subgroups remained similar (59.4% White men, 57.1% White women, 83.3% Black men, and 82.4% Black women).

COMMENT

In this study, the implementation of an efficient and accessible screening program can help to eliminate demographic inequities in lung cancer diagnosis and treatment. Our institutional lung cancer screening program was able to screen a large number of Black patients. There were increased screening rates of both Black men (12.9% vs 6.11%; $P < .0001$) and Black women (15.7% vs 6.68%; $P \leq .0001$) compared with the metropolitan area of the community. We also found that the screening program was able to provide equitable timely care to Black and White patients. It is a noteworthy that Black men had the lowest rate of lung cancer diagnosis.

At a statewide level, Ohio mirrors national disparities, with lower rates of screening and higher rates of lung cancer diagnoses for Black men (81.3 per 100,000 incident lung cancers) compared with their White counterparts (75.5 per 100,000 incident lung cancers). Additionally, according to data from the Centers for Disease Control and Prevention on lung cancer incidence from 2016 to 2020, only 6.9% of eligible patients are screened in

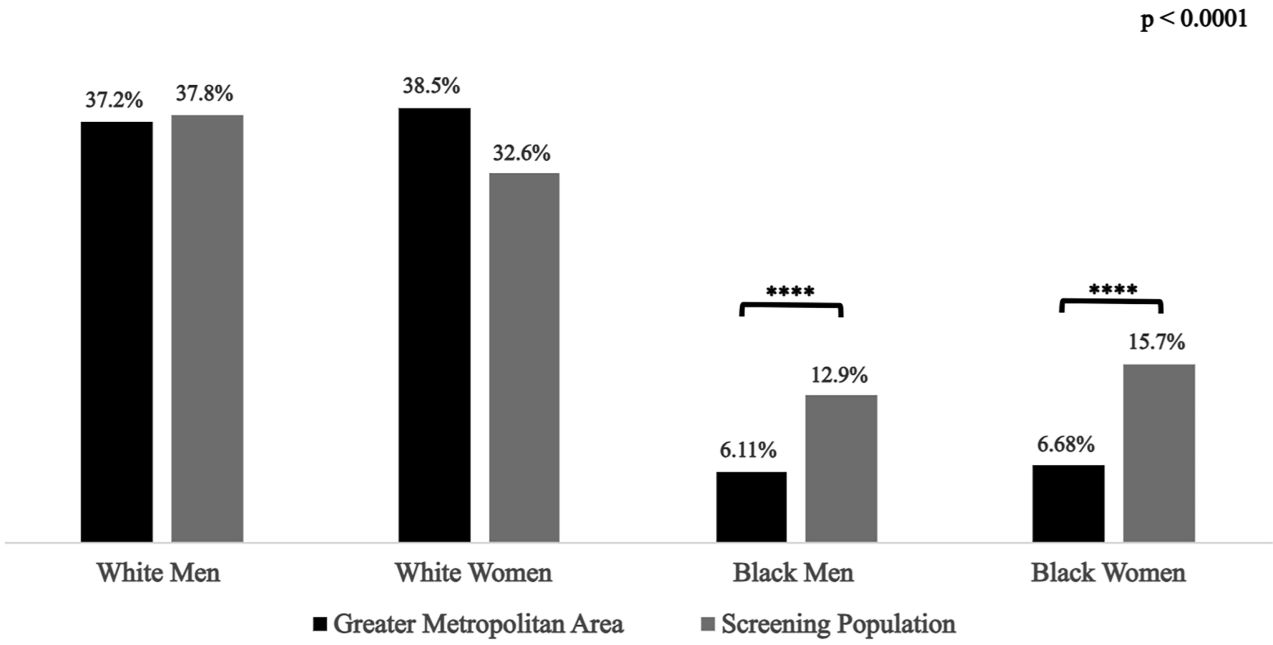


FIGURE 3 Demographic breakdown and comparison between the Cincinnati greater metropolitan area population (black shading) and the study screening population (gray shading). ****P < .0001.

Ohio.[5] However, this study demonstrated an improved rate of early-stage diagnosis for all demographics, as well as a larger proportion of Black men and women enrolled in the study screening population than in the greater metropolitan area. In essence, increasing the overall screening of high-risk racial minorities plays a role in diagnosing and treating lung cancer earlier in its progression. Our previous work on lung cancer screening use found that primary care clinics with the highest use are affiliated with training programs.[9] These clinics provide care for a higher number of racial minority patients, a factor that could contribute to the increased number of racial minority patients in the screened cohort. Additionally, our program has participated in community events to increase lung cancer screening in racial minority patients, and this also could contribute to our screening cohort.

Most enrolled patients who were identified with incident lung cancer completed the continuum of care and pursued an intervention. In the study cohort, there were no significant differences in stage at diagnosis, cancer type, or rates of resection by race. Although it has been well established that Black men have lower rates of resection and worse survival after lung cancer, this study found comparable resection rates between racial subgroups (59.4% White men, 83.3% Black men, 57.1% White women, and 82.4% Black women). However, Black men were less likely to receive a diagnosis of lung cancer compared with Black women (1.5% vs 4.7%; $P = .02$). The reasons for this difference are unclear given that a large number of Black men underwent screening for lung cancer, and the proportions of Black men and women screened were similar.

There were no significant differences in time to specialist referral, time to bronchoscopy or biopsy, time to an intervention, or survival when stratified by sex, race, insurance status, smoking status, smoking history, or pulmonary risk

TABLE 2 Comparative Lung Cancer Diagnosis and Treatment Trends

Factors	White, n = 67	Black n = 23	*P* Value
Incidence of lung cancer in screening population	3.9	3.3	.46
Age, y	68 [61–73]	66 [63–68]	.21
Sex, male	32 (47.8)	6 (26.1)	.07
Stage			.80
I	31 (53.5)	11 (50.0)	
II	13 (22.4)	6 (27.3)	
III	8 (13.8)	4 (18.2)	
IV	6 (10.3)	1 (4.6)	
Treatment			.95
Resection only	27 (43.6)	10 (45.5)	
Resection and chemotherapy	9 (14.5)	4 (18.1)	

Values are %, median [interquartile range], n (%), or *P*.

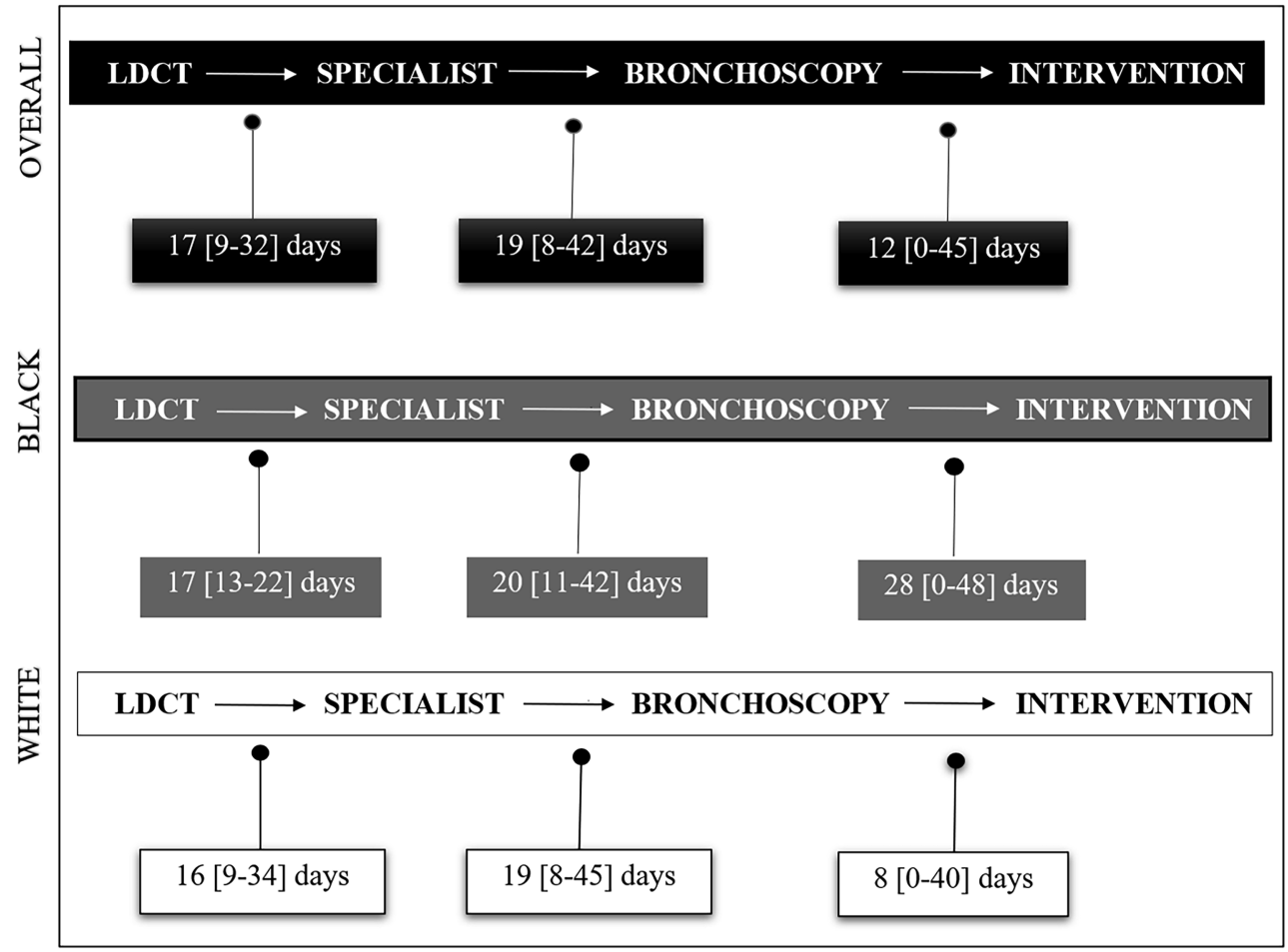


FIGURE 4 Average timeline of care from screening low-dose computed tomography (LDCT) to specialist visit, to bronchoscopy and biopsy, and to intervention (surgical resection, chemotherapy, radiation, or a combination). Overall (black shading); Black patients (gray shading); White patients (no shading).

factors. One group of patients with greater than 8 weeks from initial CT to intervention represented a small subset of patients with "delayed care." However, when this subset analyzing further, this phenomenon resulted from a slight increase in median time of each step in the workup, with no significant differences in demographics or deprivation indexes. There was no specific bottleneck identified in the process, but rather a few patients who had slower workup progression. These metrics also provide opportunities for our lung cancer screening program to improve its efficiency with quality improvement projects.

There are several limitations to consider. The study population comprised patients served by the University of Cincinnati, an institution that primarily treats both rural and large urban populations with a diverse demographic makeup. The study cohort is representative of these populations; however, it is impossible to know whether the improved screening rates for racial minorities was contributed to by the overall demographics of the cohort. Patients in the incident lung cancer cohort received diagnoses through ICD-10 codes. In the absence of individual chart reviews of the entire study group, this does not exclude the possibility that some patients with lung cancer were overlooked. Additionally, for any patients who either did not follow up or had a delayed treatment course, the study can only speculate the contributory reasons. For any number of these patients, their diagnostic and treatment courses could have been extended because of comorbidities, unforeseen medical challenges, or socioeconomic circumstances out of their control. This study is a microcosm of the complex medical journey patients with a lung cancer diagnosis undergo, and it in no way represents their entire experience.

This study highlights the true utility of establishing a lung cancer screening program in neutralizing racial and sex inequities. Through its implementation, established disparities experienced by Black male patients were eliminated. However, current evidence indicates that implementation of effective LDCT screening programs would benefit from the consideration of minority and low socioeconomic status for screening eligibility. Historically, vulnerable populations are screened at inequitable rates as a result of inaccurate smoking history, incongruent screening eligibility, low health literacy and access, and other barriers to care.[21] Further efforts must be invested in understanding how these mechanisms contribute to inequities in surgical lung cancer screening, diagnosis, treatment, and outcomes. The establishment of organized lung cancer screening programs may not be sufficient to achieve equitable care, and other interventions, including routine program appraisals and quality metrics, may be necessary. This study demonstrates a step in the right direction. However, there is a continual need to combat disparities in the treatment of lung cancer.

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DISCLOSURES

The authors have no conflicts of interest to disclose.

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